

Introduction to Optimization with Matlab

Tutorial 1 – Friday May 16, 2014

Outline

- Matlab
- Matlab Optimization Toolbox
- Optimization Problem
- Design Problems

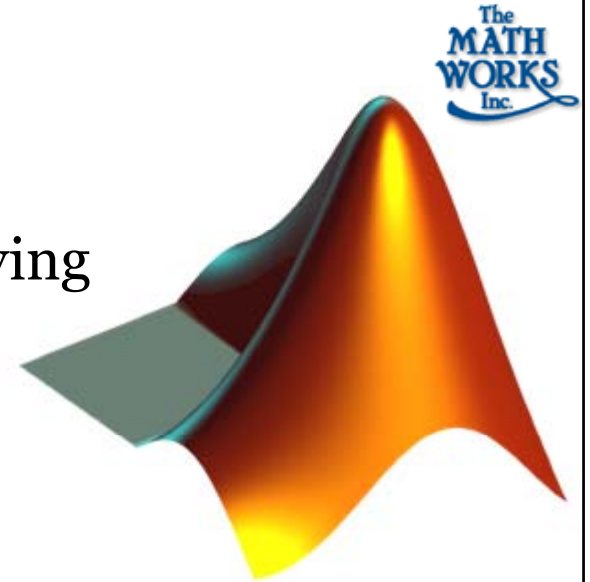
Outline

- **Matlab**
- Matlab Optimization Toolbox
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- Design Problems

Introduction to Matlab

- **What is Matlab?**

- MATLAB is a program for doing numerical computation. It was originally designed for solving linear algebra type problems using matrices.
- It's name is derived from **MA**Trix **LAB**oratory.
- MATLAB is a software environment for interactive numerical computations.
- MATLAB is a high-level language and interactive environment that enables you to perform computationally intensive tasks faster than with traditional programming languages such as C, C++, and Fortran.



Introduction to Matlab

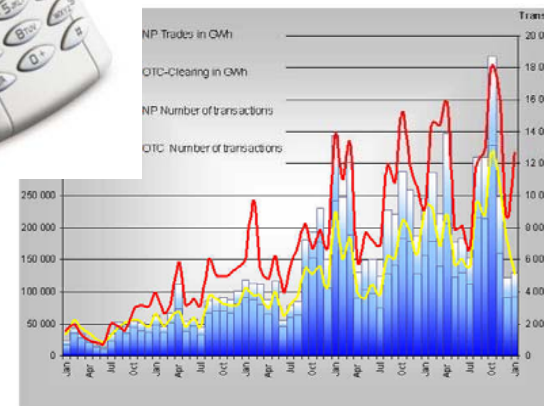
- **Tasks**

- Matrix computations and linear algebra
- Solving nonlinear equations
- Numerical solution of differential equations
- Mathematical optimization
- Statistics and data analysis
- Signal processing
- Modelling of dynamical systems
- Solving partial differential equations
- Simulation of engineering systems

Introduction to Matlab

- Usage

Matlab used (on a daily basis) in many engineering companies



Introduction to Matlab

- **Background**

Matlab = Matrix Laboratory

- Originally a user interface for numerical linear algebra routines (Lapack/Linpack)
- Commercialized 1984 by The Mathworks
- Since then heavily extended (defacto-standard)

- **Alternatives**

Matrix-X

Octave (free; GNU)

Lyme (free; Palm)

- **Complements**

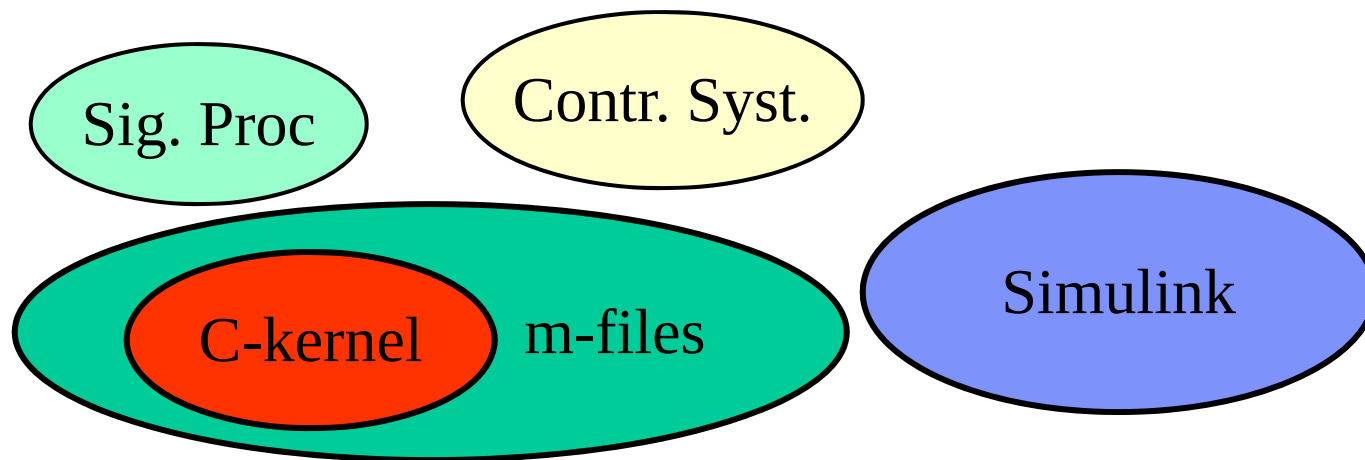
Maple (symbolic)

Mathematica (symbolic)

Introduction to Matlab

- **Background**

- Core functionality: compiled C-routines
- Most functionality is given as m-files, grouped into toolboxes
- m-files contain source code, can be copied and altered
- m-files are platform independent (PC, Unix/Linux, MAC)
- Simulation of dynamical systems is performed in Simulink.



Introduction to Matlab

- **Tutorial**

See: Matlab tutorial

HERIOT-WATT UNIVERSITY

DEPARTMENT OF COMPUTING AND ELECTRICAL
ENGINEERING. Available on the course website.

Outline

- Matlab
- **Matlab Optimization Toolbox**
- Optimization Problem
- Design Problems

Matlab Optimization Toolbox

- **What Is the Optimization Toolbox?**

- Collection of functions that extends Matlab capabilities
- Includes functions to solve difficult optimization problems
 - Minimization and Maximization Problems
 - Multi-variable objective function optimization
 - Linear and Quadratic optimization
 - Nonlinear system of equation solving
 - Nonlinear least squares and curve-fitting

Matlab Optimization Toolbox

- **Minimization**

- Unconstrained minimization function

- FMINUNC(FUN,Xo)

- Constrained minimization function

- FMINCON(FUN,Xo,A,B,Aeq,Beq,LB,UB, NONLCON)

- **Equation Solving**

- Nonlinear equation solving

- FSOLVE(FUN,Xo)

- **Curve Fitting**

- Nonlinear curve fitting

- LSQCURVEFIT(FUN,Xo,XDATA,YDATA)

Matlab Optimization Toolbox

- Examples using the Matlab Optimization Toolbox
 - Unconstrained Example
 - Constrained Example
 - Constrained With Bounds Example

Matlab Optimization Toolbox

- **Unconstrained Example**

```
function unconstrainedOptimizationExample

%clearing all previous data
clc,clear,close all

% Make a starting guess at the solution
x0 = [0,0];

% Overwriting the default optimization options
options = optimset('Display','none','LargeScale','off');

% Calling the constrained minimization function
[x,fval,exitflag,output] = fminunc(@objfun,x0,options)

%Defining the objective function
function f = objfun(x)
f = exp(x(1))*(4*x(1)^2+2*x(2)^2+4*x(1)*x(2)+2*x(2)+1);
```

```
x =

    0.5000   -1.0000

fval =

    1.8304e-015

exitflag =

     1

output =

    iterations: 7
    funcCount: 33
    stepsize: 1
firstorderopt: 2.4568e-008
    algorithm: 'medium-scale: Quasi-Newton line search'
```

Matlab Optimization Toolbox

- Constrained Example

```
function constrainedOptimizationExample

%clearing all previous data
clc,clear,close all

% Make a starting guess at the solution
x0 = [-1,1];

% Overwriting the default optimization options
options = optimset('Display','none','LargeScale','off');
% Calling the constrained minimization function
[x, fval] = fmincon(@objfun,x0,[],[],[],[],[],[],@confun,options)
confun(x)

% Defining the set of Non-linear inequalities and equalities constraints
function [c, ceq] = confun(x)
% Nonlinear inequality constraints
c = [1.5 + x(1)*x(2) - x(1) - x(2);
-x(1)*x(2) - 10];
% Nonlinear equality constraints
ceq = [];

%Defining the objective function
function f = objfun(x)
f = exp(x(1)) * (4*x(1)^2+2*x(2)^2+4*x(1)*x(2)+2*x(2)+1);
```

```
x =

    -9.5474    1.0474

fval =

    0.0236

exitflag =

     1

output =

    iterations: 8
    funcCount: 28
    lssteplength: 1
    stepsize: 4.2503e-004
    algorithm: 'medium-scale: SQP, Quasi-Newton, line-search'
    firstorderopt: 8.5125e-007
```

Matlab Optimization Toolbox

- **Constrained With Bounds Example**

```
function constrainedWithBounds

%clearing all previous data
clc,clear,close all

lb = [0,0]; % Set lower bounds
ub = [ ]; % No upper bounds
x0 = [-1,1]; % Initial guess

% Overwriting the default optimization options
options = optimset('Display','none','LargeScale','off');
% Calling the constrained minimization function
[x, fval] = fmincon(@objfun,x0,[],[],[],[],lb,ub,@confun,options)

% Defining the set of Non-linear inequalities and equalities constraints
function [c, ceq] = confun(x)
% Nonlinear inequality constraints
c = [1.5 + x(1)*x(2) - x(1) - x(2);
-x(1)*x(2) - 10];
% Nonlinear equality constraints
ceq = x(1)^2 + x(2) - 1;

%Defining the objective function
function f = objfun(x)
f = exp(x(1)) * (4*x(1)^2+2*x(2)^2+4*x(1)*x(2)+2*x(2)+1);
```

```
x =

    -0.1429    1.2563

fval =

    5.2296

exitflag =

     0

output =

    iterations: 41
    funcCount: 201
    lssteplength: 0.0039
    stepsize: 0.0300
    algorithm: 'medium-scale: SQP, Quasi-Newton, line-search'
    firstorderopt: 723.1771
```


Outline

- Matlab
- Matlab Optimization Toolbox
- **Optimization Problem**
- Design Problems

Optimization Problem

- **Statement of an Optimization Problem**

An optimization or a mathematical programming problem can be stated as follows.

$$\text{Find } \mathbf{X} = \begin{Bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{Bmatrix} \text{ which minimize / maximize } f(\mathbf{X})$$

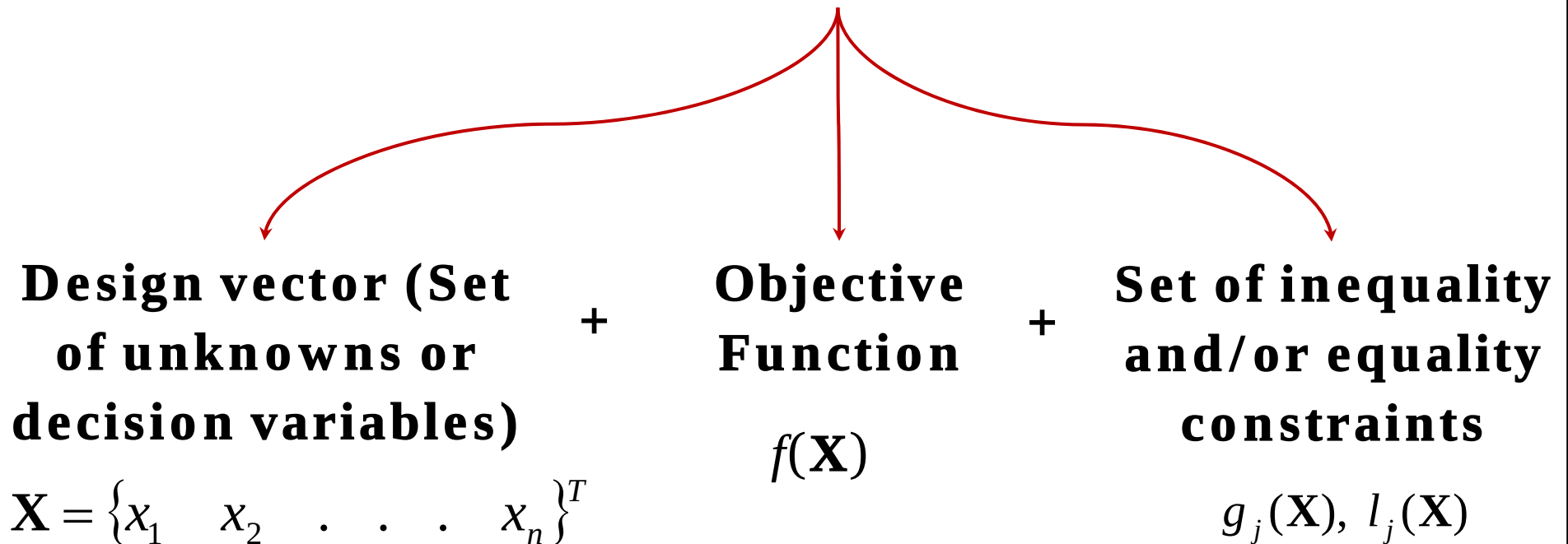
subject to the constraints

$$g_j(\mathbf{X}) \leq 0, \quad j = 1, 2, \dots, m \quad \text{Inequality constraints}$$

$$l_j(\mathbf{X}) = 0, \quad j = 1, 2, \dots, p \quad \text{Equality constraints}$$

Optimization Problem

Optimization Problem



Optimization Problem

• Product Mix Problem

Reddy Mikks produces both interior and exterior paints from two raw materials, M1 and M2. The following table provides the basic data of the problem:

	Tons of raw material per ton of		Maximum daily availability (tons)
	Exterior paint	Interior paint	
Raw material (M1)	6	4	24
Raw material (M2)	1	2	6
Profit per ton (\$1000)	5	4	

A market survey indicates that the daily demand for interior paint cannot exceed that of exterior paint by more than 1 ton. Also, the maximum daily demand of interior paint is 2 tons.

Reddy Mikks wants to determine the optimum (best) product mix of interior and exterior paints that maximizes the total daily profit.

● Decision/design variable

● Objective function

● Constraint

Optimization Problem

- **Food Mix at Farm**

Ozark Farms uses at least 800 lb of special feed daily. The special feed is a mixture of corn and soybean meal with the following composition:

Feedstuff	lb per lb of feedstuff		Cost (\$/lb)
	Protein	Fiber	
Corn	0.09	0.02	0.30
Soybean	0.60	0.06	0.90

The dietary requirements of the special feed are at least 30% protein and at most 5% fiber.

Ozark Farms wishes to determine the daily minimum-cost feed mix.

Optimization Problem

- **Fluid Gallons**

An 8-gallon jug is filled with fluid. Given two empty 5- and 3-gallon jugs, we want to divide the 8 gallons of fluid into two equal parts using the three jugs. No other measuring devices are allowed.

What is the smallest number of pourings needed to achieve this result?

Outline

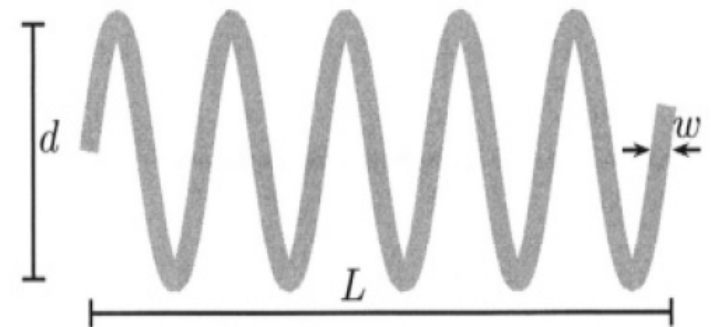
- Matlab
- Matlab Optimization Toolbox
- Optimization Problem
- **Design Problems**

Design Problem

- **Spring Design**

A very simple design problem in engineering is to design a spring under tension and/or compression for given requirements or constraints including **minimum deflection**, **outer diameter**, **frequency**, and **maximum shear stress**.

For a given problem such as forming a spring using a wire, the adjustable parameters or design variables are the **wire diameter** w (thickness), the **coil diameter** d , and the **length** L (or equivalently, the number of active coils).



Design Problem

- **Spring Design**

This design optimization problem can be formulated as follows:

$$\text{minimize } f(\mathbf{x}) = (2 + L)dw^2,$$

$$\text{subject to } g_1(\mathbf{x}) = 1 - \frac{d^2L}{7178w^4} \leq 0$$

$$g_2(\mathbf{x}) = \frac{4d^2 - wd}{12566dw^3 - w^4} + \frac{1}{5108w^2} - 1 \leq 0$$

$$g_3(\mathbf{x}) = 1 - \frac{140.45w}{d^2L} \leq 0$$

$$g_4(\mathbf{x}) = \frac{w+L}{1.5} - 1 \leq 0$$

The simple bounds are

$$0.05 \leq w \leq 2.0, \quad 0.25 \leq d \leq 1.3, \quad 2.0 \leq L \leq 15.0$$

If we use the Matlab program, we can find the following best solution:

$$f_* \approx 0.0075$$

$$\mathbf{x}_* \approx (0.0500, 0.2500, 9.9877)$$

[1]

Design Problem

- **Spring Design**

$$\mathbf{X} = \begin{Bmatrix} w \\ d \\ L \end{Bmatrix} = \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} \quad f(\mathbf{X}) = (2 + x_3)x_2x_1^2$$

$$g_1(\mathbf{X}) = 1 - \frac{x_2^3 x_3}{7178 x_1^4} \leq 0$$

$$g_2(\mathbf{X}) = \frac{4x_2^2 - x_1 x_2}{12566 x_2 x_1^3 - x_1^4} + \frac{1}{5108 x_1^2} - 1 \leq 0$$

$$g_3(\mathbf{X}) = 1 - \frac{140.45 x_1}{x_2^2 x_3} \leq 0$$

$$g_4(\mathbf{X}) = \frac{x_1 + x_2}{1.5} - 1 \leq 0$$

$$\text{find } \mathbf{X}^* = \arg \min_{\mathbf{X}} f(\mathbf{X})$$

$$\text{s.t. } g_i(\mathbf{X}) \leq 0 \quad i = 1, 2, 3, 4$$

Design Problem

- **Spring Design**

The bounds can be written as $Lb \leq x \leq Ub$

with the lower bound $Lb = [0.05; 0.25; 2.0]$

and the upper bound $Ub = [2.0; 1.3; 15.0]$

Starting from an educated guess $x_0 = [0.1; 0.5; 10]$

Design Problem

• Spring Design

```
% Spring Design Optimization using Matlab fmincon
function spring
x0=[0.1; 0.5; 10];
Lb=[0.05; 0.25; 2.0]; Ub=[2.0; 1.3; 15.0];

% call matlab optimization toolbox
[x,fval]=fmincon(@objfun,x0, [],[],[], [], Lb,Ub,@nonfun)

% Objective function
function f=objfun(x)
f=(2+x(3))*x(1)^2*x(2);

% Nonlinear constraints
function [g,geq]=nonfun(x)

% Inequality constraints
g(1)=1-x(2)^3*x(3)/(7178*x(1)^4);
gtmp=(4*x(2)^2-x(1)*x(2))/(12566*x(2)*x(1)^3-x(1)^4);
g(2)=gtmp+1/(5108*x(1)^2)-1;
g(3)=1-140.45*x(1)/(x(2)^2*x(3));
g(4)=x(1)+x(2)-1.5;

% Equality constraints [none]
geq= [] ;
```

Run

```
x =
    0.0500
    0.2845
    2.0000
fval =
    0.0028
```

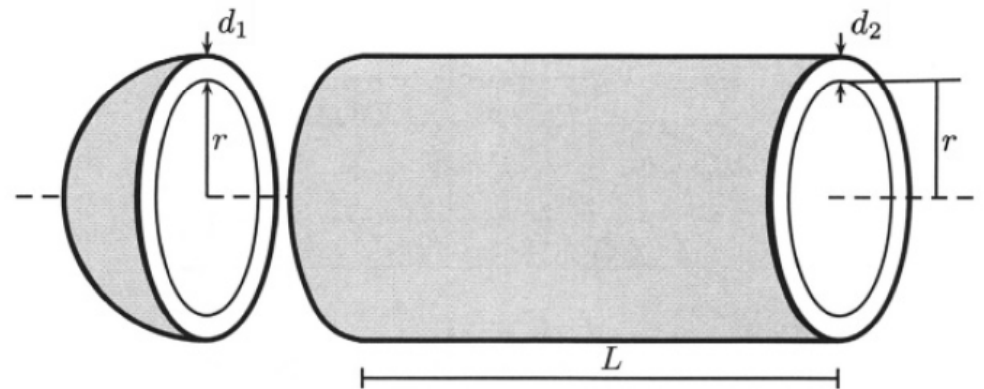
Design Problem

- **Pressure Vessels**

Pressure vessels are literally everywhere such as bottles of sparkling drink, and gas tanks. For a given volume and working pressure, the basic aim of designing a cylindrical vessel is to **minimize the total cost**.

Typically, the design variables are:

- ◇ the thickness d_1 of the head,
- ◇ the thickness d_2 of the body,
- ◇ the inner radius r , and
- ◇ the length L of the cylindrical section.



Design Problem

- **Pressure Vessels**

This is a well-known test problem for and it can be written as

$$\text{minimize } f(\mathbf{x}) = 0.6224d_1rL + 1.7781d_2r^2 + 3.1661d_1^2L + 19.84d_1^2r,$$

$$\begin{aligned} \text{subject to } g_1(\mathbf{x}) &= -d_1 + 0.0193r \leq 0 \\ g_2(\mathbf{x}) &= -d_2 + 0.00954r \leq 0 \\ g_3(\mathbf{x}) &= -\pi r^2L - \frac{4\pi}{3}r^3 + 1296000 \leq 0 \\ g_4(\mathbf{x}) &= L - 240 \leq 0. \end{aligned}$$

The simple bounds are $0.0625 \leq d_1, d_2 \leq 99 \times 0.0625$,

$$10.0 \leq r, \quad L \leq 200.0.$$

It is worth pointing out that d_1 and d_2 should only take discrete values of integer multiples of 0.0625.

Solution: $f_* = 6059.714$,

$$\mathbf{x}_* = (0.8125, 0.4375, 42.0984, 176.6366)$$

[1]

Design Problem

- **Pressure Vessels**

$$\mathbf{X} = (d_1 \quad d_2 \quad r \quad L)^T = (x_1 \quad x_2 \quad x_3 \quad x_4)^T$$

minimize

$$f(\mathbf{X}) = 0.6224x_1x_3x_4 + 1.7781x_2x_3^2 + 3.1661x_1^2x_4 + 19.84x_1^2x_3$$

subject to

$$g_1(\mathbf{X}) = -x_1 + 0.0193x_3 \leq 0$$

$$g_2(\mathbf{X}) = -x_2 + 0.00954x_3 \leq 0$$

$$g_3(\mathbf{X}) = x_4 - 240 \leq 0$$

$$g_4(\mathbf{X}) = -\pi x_3^2 x_4 - \frac{4\pi}{3} x_3^3 + 1296000 \leq 0$$

Design Problem

- **Pressure Vessels**

The first three inequalities g_1, g_2, g_3 can be written as

$$\mathbf{A}x \leq b$$

$$\mathbf{A} = \begin{pmatrix} -1 & 0 & 0.0193 & 0 \\ 0 & -1 & 0.00954 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad b = \begin{pmatrix} 0 \\ 0 \\ 240 \end{pmatrix}$$

The simple bounds can be rewritten as

$$\text{Lb} = [d; d; 10; 10]; \text{Ub} = [99 * d; 99 * d; 200; 200],$$

where $d = 0.0625$.

Design Problem

• Pressure Vessels

```
% Design Optimization of a Pressure Vessel
function pressure_vessel
x0=[1; 1; 20; 50];
d=0.0625;

% Linear inequality constraints
A=[-1 0 0.0193 0;0 -1 0.00954 0;0 0 0 1];
b=[0; 0; 240];

% Simple bounds
Lb=[d; d; 10; 10]; Ub=[99*d; 99*d; 200; 200];
options=optimset('Display','iter','TolFun',1e-08);
[x,fval]=fmincon(@objfun,x0,A,b, [], [], Lb,Ub,@nonfun,options)

% The objective function
function f=objfun(x)
f=0.6224*x(1)*x(3)*x(4)+1.7781*x(2)*x(3)^2+3.1661*x(1)^2*x(4)+19.84*x(1)^2*x(3)

% Nonlinear constraints
function [g,geq]=nonfun(x)

% Nonlinear inequality
g=-pi*x(3)^2*x(4)-4*pi/3*x(3)^3+1296000;

% Equality constraint [none]
geq= [];
```

Run

```
x =
    0.7782
    0.3846
   40.3196
   200.0000
fval =
   5885.3328
```

References

1. Xin-SheYang. *Engineering Optimization: An Introduction with Metaheuristic Applications*. John Wiley & Sons, 2010.